

Supplementary Material

Supplementary Table 1. Description of C-BARQ subscales and miscellaneous items.

C-BARQ scale	Questions per scale	Description
Excitability	2	Strong reactions to arousing events, e.g., being taken on a walk or on a car trip.
Stranger-directed aggression	3	Aggressive behavior/s towards unfamiliar people while in the home, outside/ yard or on a leash.
Owner-directed aggression	3	Aggressive behaviors towards owner when approached while eating or in possession of other high value objects.
Dog-directed aggression	2	Aggressive behaviors toward unfamiliar dogs.
Dog rivalry (familiar dog aggression)	2	Aggressive behaviors towards household dogs.
Stranger-directed fear	2	Fearful behaviors when approached by unfamiliar people.
Nonsocial fear	3	Fearful behaviors in response to loud noises, unfamiliar objects, and unfamiliar situations.
Dog-directed fear	2	Fearful behaviors towards unfamiliar dogs.
Touch sensitivity	2	Fearful behaviors when having nails clipped, being groomed, or bathed by a household member.
Separation-related behavior	3	Signs of anxiety, vocalization or destructive behavior while separated from the owner.
Attachment/attention-seeking	2	Tendency to solicit attention and maintain close proximity to owner.
Training difficulty	3	Tendency to become easily distracted or ignore owner commands.
Chasing	2	Tendency to chase small animals, e.g., birds, squirrels, rabbits.

Energy	2	Playful, boisterous, active, and always on the go.
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Supplementary Table 2. Number of valid cases for each mini C-BARQ subscale or miscellaneous item.

	Pet dogs	Relinquished dogs
C-BARQ scale		
Excitability	416	416
Stranger-directed aggression	399	420
Owner-directed aggression	417	421
Dog-directed aggression	385	400
Dog rivalry (familiar dog aggression)	340	323
Stranger-directed fear	410	408
Nonsocial fear	402	416
Dog-directed fear	380	405
Touch sensitivity	357	340
Separation-related behavior	417	414
Attachment/attention-seeking	423	418
Training difficulty	406	420
Chasing	391	410
Energy	426	423
C-BARQ miscellaneous items		
Chews inappropriate objects	425	421
Pulls on leash	424	415
Persistent barking	421	422
Tail chasing	423	421
Urine marking	423	420
Urines when left alone	419	410
Defecates when alone	419	406
Hyperactive	424	422
Escapes from home/yard	399	418

Supplementary Table 3. Dog breeds included in the study.

	Pet dogs	Relinquished dogs	Total
Breed			
Airedale Terrier	2	2	4
Akita	11	11	22
Alaskan Malamute	1	1	2

American Bulldog	5	5	10
American Pit Bull Terrier	14	14	28
American Staffordshire Terrier	1	1	2
Australian Cattle Dog	8	8	16
Australian Shepherd	7	7	14
Basenji	1	1	2
Basset Hound	1	1	2
Beagle	14	14	28
Bearded Collie	1	1	2
Bichon Frise	3	3	6
Border Collie	9	9	18
Border Terrier	1	1	2
Boston Terrier	3	3	6
Boxer	17	17	34
Bull Terrier	1	1	2
Cairn Terrier	1	1	2
Cane Corso	1	1	2
Chihuahua	16	16	32
Chow Chow	9	9	18
Cocker Spaniel	6	6	12
Collie (Smooth)	3	3	6
Coonhound	2	2	4
Corgi	3	3	6
Curr	2	2	4
Dachshund	3	3	6
Doberman Pinscher	1	1	2
English Springer Spaniel	1	1	2
French Bulldog	1	1	2
German Shepherd	32	32	64
German Shorthaired Pointer	1	1	2
Golden Retriever	6	6	12
Great Dane	2	2	4
Greyhound	1	1	2
Havanese	1	1	2
Hound mix	4	4	8
Irish Setter	1	1	2
Italian Greyhound	2	2	4
Jack Russell Terrier	6	6	12

Keeshond	2	2	4
Labrador Retriever	84	84	168
Lhasa Apso	5	5	10
Manchester Terrier	2	2	4
Mastiff (English)	2	2	4
Miniature Pinscher	3	3	6
Mixed Breed/ Unknown	5	5	10
Neapolitan Mastiff	3	3	6
Newfoundland	1	1	2
Parson Russell Terrier	5	5	10
Pharaoh Hound	1	1	2
Pit Bull	37	37	74
Pit Bull mix	6	6	12
Pointer	2	2	4
Pomeranian	3	3	6
Poodle	10	10	20
Pug	6	6	12
Rat Terrier	1	1	2
Rhodesian Ridgeback	2	2	4
Rottweiler	10	10	20
Saint Bernard	1	1	2
Schipperke	1	1	2
Schnauzer	4	4	8
Shar Pei	2	2	4
Shetland Sheepdog	1	1	2
Shih Tzu	6	6	12
Siberian Husky	9	9	18
Weimaraner	2	2	4
West Highland White Terrier	2	2	4
Whippet	1	1	2
Yorkshire Terrier	3	3	6
Total	427	427	854